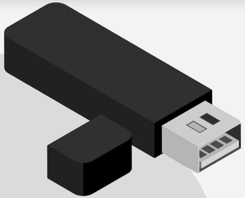
Secure Environment for Academic Assessments

An isolated setup in the palm of your hand

Diogo Silva, Duarte Santos, Gabriel Janicas, Miguel Reis, Rodrigo Moço, Tiago Garcia



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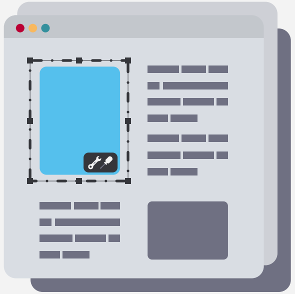
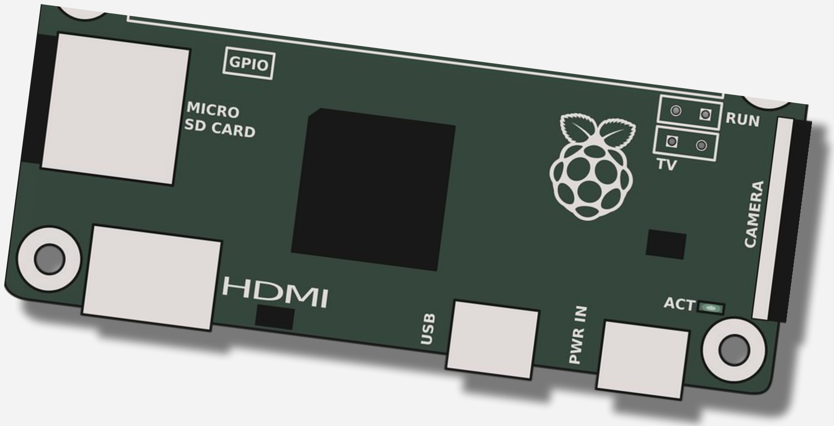
Problem





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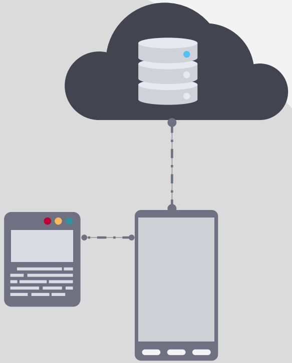
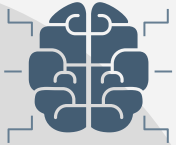
GOALS

Goals



Main goal

- ➔ Create a Plug and Play alternative to SEB, making it available on all PC devices without the need to install any extra software.



Extra goals

- ➔ Create a system to enable better control over the examinees;
- ➔ Replicate and ship the final product to the rest of the university for general usage.

03

Expected Results

Expected Results

- Finish with a commerciable and distributable product to offer its user (the examinee) a fully operational system designed to academic assessment;
- The examiner will be able to monitor the examinee's activity through a webapp, in a safe private network with no access to the internet;
- Allow the examinee to use its personal computer instead of the academic institution's ones, allowing for a much better and fine-tuned experience, always guaranteeing there has no way to access the internet or any other way of cheating.

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Limitations

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Team and Communication

Team members

Miguel Reis

Duarte Santos

Tiago Garcia

Gabriel Janicas

Diogo Silva

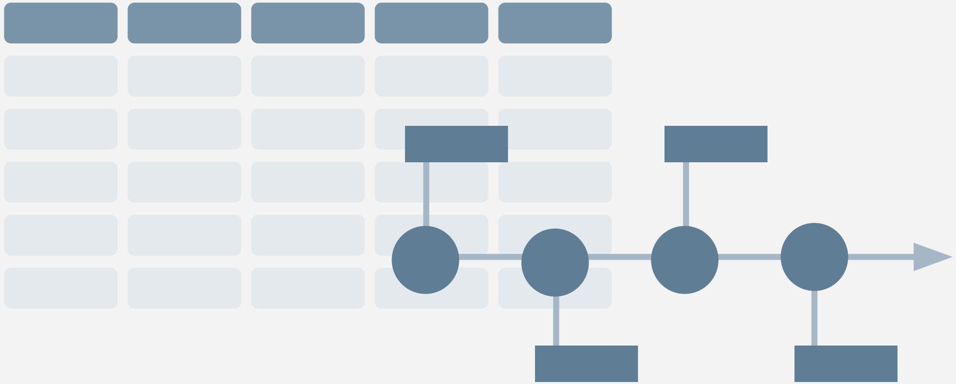
Rodrigo Moço

Team supervisors

João Almeida

João Paulo Barraca

Communication



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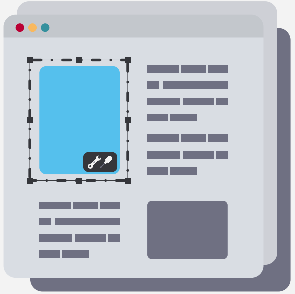
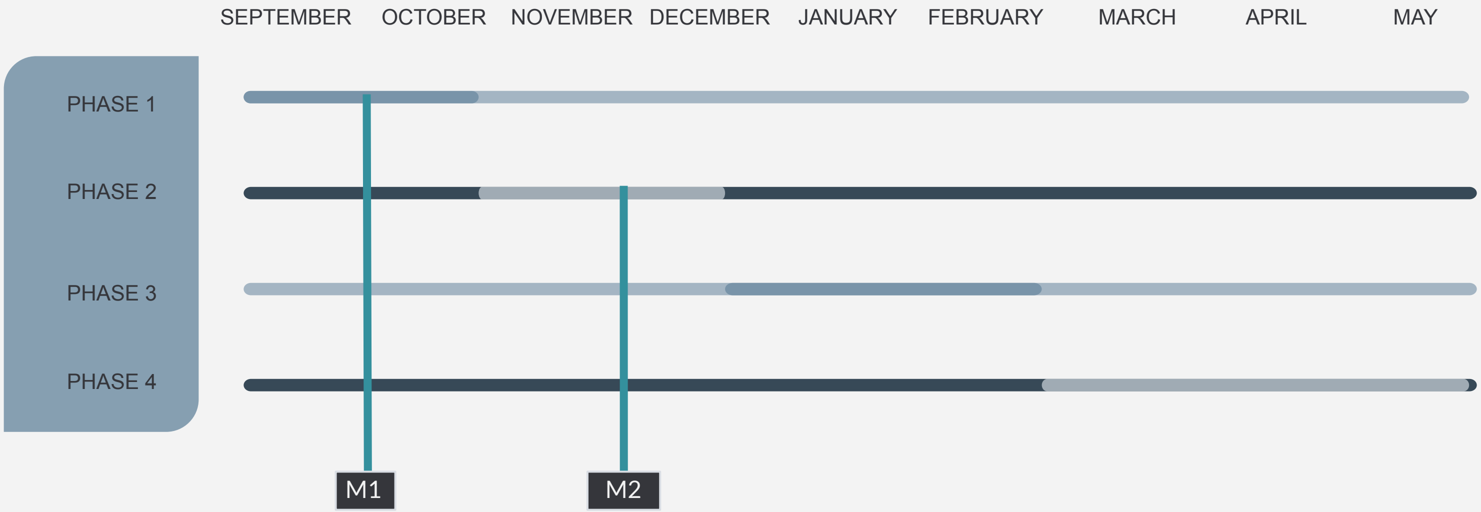
Calendar and Tasks



Calendar


SEPTEMBER
OCTOBER
NOVEMBER
DECEMBER
JANUARY
FEBRUARY
MARCH
APRIL
MAY
PHASE 1
PHASE 2
PHASE 3
PHASE 4
M1
M2

TASK LIST

State of Art	Requirements	Security & Isolation	Development	Documentation & Usability
Research similar projects and solutions	Hardware	Set requirements for secure access and authentication	Develop OS and software	Document system architecture and design
Identify and evaluate current solutions for exam environments	Identify software and tools necessary for exams	Identify potential security threats	Configure devices, testing functionality and performance	Test interface and system for improvements
Research and compare high-performance technologies	Determine monitoring tools	Establish data protection measure	Create User interface	Document deployment procedures for exam staff
Examine OS hardening techniques	System Architecture and Communication		Create a system for distributing exams and resources	Milestones

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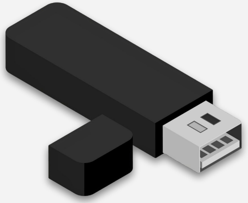
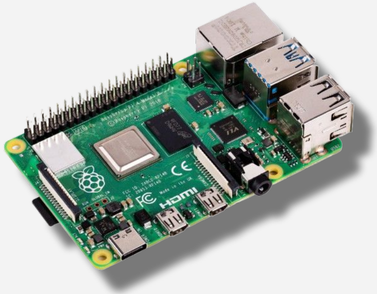
Resources



Resources

- PI Zero 2 W
- Raspberry Pi Model B 4GB
- Memory Card microSD
- Cooler Pi Zero 2



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Website

https://portableisolatedenvironment.github.io/

THANKS!

Do you have any questions?